



Qualification Pack

Aquaponics Cultivator

QP Code: CUTM/AGR/Q0701

Version: 1.0

NSQF Level: 4

Centurion University of Technology and Management || Aluri Nagar, Paralakhemundi, R Sitapur
Uppalada || email:partha@cutm.ac.in



Qualification Pack

Contents

CUTM/AGR/Q0701: Aquaponics Cultivator	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
CUTM/AGR/N0701: Introduction to Aquaponics	5
CUTM/AGR/N0702: Aquaponics System Layout and its Components.	9
CUTM/AGR/N0703: The Nitrogen cycle/Water testing and Quality control	13
CUTM/AGR/N0704: Ornamental fish in Aquaponics system	17
CUTM/AGR/N0705: Herbaceous Plants or Hydrophytes in Aquaponics system	21
CUTM/VSQ/N7803: Maintain Health and Safety at Work Place.	25
DGT/VSQ/N0102: Employability Skills (60 Hours)	30
Assessment Guidelines and Weightage	37
<i>Assessment Guidelines</i>	37
<i>Assessment Weightage</i>	38
Acronyms	40
Glossary	41



Qualification Pack

CUTM/AGR/Q0701: Aquaponics Cultivator

Brief Job Description

The Individual is responsible to understand that Aquaponics is a productive, innovative and sustainable approach to food production that combines hydroponics the practice of growing plants in water with aquaculture, the rearing of aquatic animals. Fish faces serve as a natural fertilizer for the plants in this recirculating system.

Personal Attributes

An Aquaponics Cultivator should be observant, environmentally conscious, and technically skilled. They must possess patience, problem-solving abilities, and a hands-on approach to managing integrated systems. A willingness to learn, adaptability to changing conditions, and strong attention to hygiene, water quality, and plant-fish health are essential for sustainable operations.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/AGR/N0701: Introduction to Aquaponics](#)
2. [CUTM/AGR/N0702: Aquaponics System Layout and its Components.](#)
3. [CUTM/AGR/N0703: The Nitrogen cycle/Water testing and Quality control](#)
4. [CUTM/AGR/N0704: Ornamental fish in Aquaponics system](#)
5. [CUTM/AGR/N0705: Herbaceous Plants or Hydrophytes in Aquaponics system](#)
6. [CUTM/VSQ/N7803: Maintain Health and Safety at Work Place.](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Horticulture
Occupation	Cultivator
Country	India



Qualification Pack

NSQF Level	4
Credits	17
Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	12th grade Pass with 1 Year of experience OR 10th grade pass with 3 Years of experience OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	21/10/2027
NSQC Approval Date	22/10/2024
Version	1.0
Reference code on NQR	QG-04-AG-03351-2024-V1-CUTM
NQR Version	1.0

Qualification Pack

CUTM/AGR/N0701: Introduction to Aquaponics

Description

This NOS covers the fundamental understanding of aquaponics, an integrated system combining aquaculture and hydroponics. It involves knowledge of biological processes, system setup, water quality, and environmental management. Learners gain insights into sustainable food production using closed-loop systems that are efficient, eco-friendly, and suitable for small to medium-scale cultivation.

Scope

The scope covers the following :

- The scope includes the identification and setup of basic aquaponics systems, understanding the interaction between fish, plants, and microbes, and monitoring environmental conditions. It also covers daily system maintenance, recognizing and resolving basic issues, and ensuring proper care of aquatic and plant life in an integrated and sustainable farming system.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** History, evolution and principles of Aquaponics.
- PC2.** Components of an Aquaponics system.
- PC3.** Types of an Aquaponics system.
- PC4.** Water quality monitoring.
- PC5.** Aquaponics vs traditional agriculture.
- PC6.** Future trends and Innovations of Aquaponics.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic principles and concepts of aquaponics and its sustainable advantages.
- KU2.** the symbiotic relationship between fish, plants, and bacteria in a closed-loop system.
- KU3.** the different types of aquaponics systems (media bed, NFT, deep water culture).
- KU4.** the water chemistry parameters (pH, ammonia, nitrites, nitrates) critical for system health.
- KU5.** suitable fish species and plant varieties for aquaponics.
- KU6.** basic biological processes such as the nitrogen cycle.
- KU7.** the components of an aquaponics setup: fish tank, grow beds, biofilters, and aeration systems.
- KU8.** environmental requirements: light, temperature, and humidity for optimal plant and fish growth.
- KU9.** common challenges in aquaponics systems and basic troubleshooting techniques.
- KU10.** sustainable and organic practices in aquaponics cultivation.



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** to set up a small-scale aquaponics unit following given guidelines.
- GS2.** Monitor water parameters and identify imbalances using simple testing tools.
- GS3.** Communicate effectively with team members regarding operational issues and daily tasks.
- GS4.** Follow standard operating procedures (SOPs) for system maintenance.
- GS5.** Observe and document the health of fish and plants regularly.
- GS6.** Apply basic troubleshooting skills to correct system imbalances.
- GS7.** Maintain cleanliness and hygiene in the aquaponics environment.
- GS8.** Manage time effectively to handle multiple components of the system.
- GS9.** Demonstrate responsibility in handling fish and plant resources with care.
- GS10.** Record observations and data related to plant/fish growth for future improvement.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. History, evolution and principles of Aquaponics.	4	5	-	5
PC2. Components of an Aquaponics system.	6	4	-	5
PC3. Types of an Aquaponics system.	5	9	-	5
PC4. Water quality monitoring.	5	9	-	5
PC5. Aquaponics vs traditional agriculture.	5	6	-	5
PC6. Future trends and Innovations of Aquaponics.	5	7	-	5
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0701
NOS Name	Introduction to Aquaponics
Sector	Agriculture
Sub-Sector	
Occupation	Cultivator
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQF Clearance Date	22/10/2024

Qualification Pack

CUTM/AGR/N0702: Aquaponics System Layout and its Components.

Description

This NOS outlines the knowledge and skills required to design and install the layout of an aquaponics system. It includes understanding system types, identifying components, planning spatial arrangements, and assembling all parts to ensure smooth circulation and effective functioning. It prepares learners to set up functional and scalable aquaponics units.

Scope

The scope covers the following :

- The scope of this NOS includes identifying and arranging essential components of an aquaponics system, interpreting layout plans, and ensuring proper connectivity. It focuses on structural setup, water flow design, and component placement to support healthy fish and plant growth. It also addresses safety, troubleshooting, and basic system testing.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Aquaponic System Components.
- PC2.** Circulation and Water Flow
- PC3.** Media-Based Systems
- PC4.** Nutrient Film Technique (NFT) Systems
- PC5.** Deep Water Culture (DWC) Systems
- PC6.** Hybrid Systems
- PC7.** Energy Efficiency in System Design.
- PC8.** Cost and Budgeting

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the fundamental design principles of various aquaponics systems.
- KU2.** key components of the system: fish tank, grow bed, sump tank, water pump, biofilter, and aerators.
- KU3.** the function and interdependence of each component in maintaining a healthy system.
- KU4.** plumbing layouts for efficient water circulation and nutrient flow.
- KU5.** system sizing based on space, fish stock, and plant load.
- KU6.** materials and tools required for system construction and assembly.
- KU7.** the role of mechanical and biological filtration in system health.
- KU8.** how to position components for easy access and optimal sunlight exposure for plants.
- KU9.** safety and ergonomic considerations in system layout.

Qualification Pack

KU10. potential system bottlenecks and common layout errors to avoid.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** to read and interpret simple aquaponics layout diagrams and blueprints.
- GS2.** Measure and mark physical space for installation of tanks, beds, and piping.
- GS3.** Assemble system components following layout guidelines.
- GS4.** Connect and test plumbing for leak-proof water circulation.
- GS5.** Monitor water flow rates and make necessary adjustments.
- GS6.** Troubleshoot minor structural or flow-related issues in the setup.
- GS7.** Maintain system cleanliness and ensure component alignment.
- GS8.** Work in coordination with others during installation or modification tasks.
- GS9.** Follow safe handling practices while using tools and assembling parts.
- GS10.** Maintain records of component specifications and installation procedures.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Aquaponic System Components.	5	5	-	-
PC2. Circulation and Water Flow	5	10	-	-
PC3. Media-Based Systems	5	10	-	-
PC4. Nutrient Film Technique (NFT) Systems	5	5	-	-
PC5. Deep Water Culture (DWC) Systems	5	5	-	-
PC6. Hybrid Systems	5	10	-	-
PC7. Energy Efficiency in System Design.	5	10	-	-
PC8. Cost and Budgeting	5	5	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0702
NOS Name	Aquaponics System Layout and its Components.
Sector	Agriculture
Sub-Sector	
Occupation	Cultivator
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQF Clearance Date	22/10/2024

Qualification Pack

CUTM/AGR/N0703: The Nitrogen cycle/Water testing and Quality control

Description

This NOS focuses on understanding the nitrogen cycle and conducting water quality testing in an aquaponics system. It covers the importance of maintaining optimal water parameters for fish and plant health, the function of beneficial bacteria, and the use of test kits to ensure a safe and productive environment.

Scope

The scope covers the following :

- The scope of this NOS includes monitoring and managing the nitrogen cycle through regular water testing. It involves understanding biological filtration, interpreting test results, and ensuring safe nitrogen levels. It also addresses preventive and corrective actions to maintain high-quality water, ensuring healthy aquaponics system performance.

Elements and Performance Criteria

Understanding the different quality parameters associated with aquaponics

To be competent, the user/individual on the job must be able to:

- PC1.** Important water quality parameters, ensuring optimal conditions for both fish and plant health.
- PC2.** Factors influencing bacterial populations.

Knowledge on various methods of assessment and maintenance of quality

To be competent, the user/individual on the job must be able to:

- PC3.** Nutrient balance to maintain the quality of water.
- PC4.** Fish and Plant health Assessment.
- PC5.** Management of water quality parameters within prescribed limits.
- PC6.** Assessment of Fish and plant health.
- PC7.** Quality certification standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the nitrogen cycle and its relevance in aquaponics (ammonia, nitrites, nitrates).
- KU2.** the role of beneficial bacteria in converting fish waste into plant nutrients.
- KU3.** the signs and consequences of ammonia or nitrite toxicity in fish.
- KU4.** optimal water parameters: pH, temperature, DO (Dissolved Oxygen), ammonia, nitrite, nitrate.
- KU5.** how imbalances in the nitrogen cycle affect plant and fish health.
- KU6.** with biological filtration and its importance in nitrogen processing.

Qualification Pack

- KU7.** the safe levels of nitrogen compounds for various fish species.
- KU8.** the importance of water testing frequency and record maintenance.
- KU9.** how to prepare and store water samples correctly.
- KU10.** Awareness of environmental and operational factors affecting water quality (feed, temperature, system loading).

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Perform routine water quality testing using test kits or digital meters.
- GS2.** Accurately measure and record pH, ammonia, nitrite, nitrate, and DO levels.
- GS3.** Interpret test results and identify trends or abnormalities.
- GS4.** Take corrective measures like water changes, adjusting feeding, or using biofilters.
- GS5.** Maintain calibration and cleanliness of water testing tools.
- GS6.** Apply safe practices when handling chemical test solutions.
- GS7.** Communicate water quality findings to supervisors or team members effectively.
- GS8.** Follow standard operating procedures for quality control.
- GS9.** Maintain accurate logs of water test data for system monitoring.
- GS10.** Coordinate with team members to implement timely corrective actions.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding the different quality parameters associated with aquaponics</i>	15	20	-	-
PC1. Important water quality parameters, ensuring optimal conditions for both fish and plant health.	7	10	-	-
PC2. Factors influencing bacterial populations.	8	10	-	-
<i>Knowledge on various methods of assessment and maintenance of quality</i>	25	40	-	-
PC3. Nutrient balance to maintain the quality of water.	5	5	-	-
PC4. Fish and Plant health Assessment.	5	10	-	-
PC5. Management of water quality parameters within prescribed limits.	5	5	-	-
PC6. Assessment of Fish and plant health.	5	10	-	-
PC7. Quality certification standards.	5	10	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0703
NOS Name	The Nitrogen cycle/Water testing and Quality control
Sector	Agriculture
Sub-Sector	
Occupation	Cultivator
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

Qualification Pack

CUTM/AGR/N0704: Ornamental fish in Aquaponics system

Description

This NOS covers the knowledge and practices required to rear and maintain ornamental fish in an aquaponics system. It includes fish selection, care, feeding, health management, and understanding their role in nutrient cycling and system aesthetics, thereby contributing to the sustainability and market appeal of aquaponics setups.

Scope

The scope covers the following :

- The scope involves selecting suitable ornamental fish, ensuring their optimal health and water quality, monitoring for disease, and implementing proper feeding and handling practices. It also includes understanding their role in ecosystem balance and leveraging ornamental fish to enhance the commercial and aesthetic value of aquaponics systems.

Elements and Performance Criteria

Knowledge on taxonomy of ornamental fish

To be competent, the user/individual on the job must be able to:

- PC1.** Selection of suitable ornamental fish species for aquaponics.
- PC2.** Assessing the compatibility of Ornamental fish with that of the plants.
- PC3.** Enhancing the well-being of fish in aquaponics environments.

Information about handling of ornamental fish waste

To be competent, the user/individual on the job must be able to:

- PC4.** Bio filtration and Nitrification.
- PC5.** Integrating grow beds as biological filters.
- PC6.** Designing efficient siphon or drain systems to remove excess waste.
- PC7.** Monitoring water quality parameters.

Knowledge on Waste-to-Energy Concepts:

To be competent, the user/individual on the job must be able to:

- PC8.** Exploring the potential for converting fish waste into energy.
- PC9.** Adjusting waste management strategies based on water quality results.
- PC10.** Continuous improvement strategies for minimizing waste.
- PC11.** Continuous improvement strategies for minimizing waste production.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the role of ornamental fish in aquaponics systems for aesthetic and commercial purposes.
- KU2.** commonly used ornamental fish species suitable for aquaponics (e.g., goldfish, koi, guppies).

Qualification Pack

- KU3.** the optimal environmental conditions for ornamental fish health (temperature, pH, DO, ammonia levels).
- KU4.** dietary requirements and feeding patterns specific to ornamental fish.
- KU5.** common diseases and stress indicators in ornamental fish and preventive care techniques.
- KU6.** quarantine procedures before introducing new ornamental fish to the system.
- KU7.** the symbiotic relationship between ornamental fish, beneficial bacteria, and plants.
- KU8.** the impact of fish behavior on system balance and plant health.
- KU9.** the regulatory and ethical guidelines for handling ornamental fish.
- KU10.** breeding basics for ornamental fish in a controlled aquaponics environment.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Monitor the health and behavior of ornamental fish regularly.
- GS2.** Feed ornamental fish using appropriate quantities and intervals.
- GS3.** Maintain cleanliness in tanks to prevent stress and disease in fish.
- GS4.** Handle ornamental fish carefully during transfers or inspections.
- GS5.** Identify early signs of stress, disease, or injury in ornamental fish.
- GS6.** Perform basic water testing to ensure optimal living conditions.
- GS7.** Document fish health, feeding routines, and water quality metrics.
- GS8.** Implement biosecurity practices to avoid cross-contamination.
- GS9.** Communicate with supervisors or aquaponics experts about fish health issues.
- GS10.** Follow safety and hygiene protocols during daily operations.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Knowledge on taxonomy of ornamental fish</i>	9	25	-	-
PC1. Selection of suitable ornamental fish species for aquaponics.	3	10	-	-
PC2. Assessing the compatibility of Ornamental fish with that of the plants.	3	5	-	-
PC3. Enhancing the well-being of fish in aquaponics environments.	3	10	-	-
<i>Information about handling of ornamental fish waste</i>	10	20	-	-
PC4. Bio filtration and Nitrification.	3	5	-	-
PC5. Integrating grow beds as biological filters.	2	5	-	-
PC6. Designing efficient siphon or drain systems to remove excess waste.	2	5	-	-
PC7. Monitoring water quality parameters.	3	5	-	-
<i>Knowledge on Waste-to-Energy Concepts:</i>	11	25	-	-
PC8. Exploring the potential for converting fish waste into energy.	5	5	-	-
PC9. Adjusting waste management strategies based on water quality results.	2	5	-	-
PC10. Continuous improvement strategies for minimizing waste.	2	10	-	-
PC11. Continuous improvement strategies for minimizing waste production.	2	5	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0704
NOS Name	Ornamental fish in Aquaponics system
Sector	Agriculture
Sub-Sector	
Occupation	Cultivator
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

Qualification Pack

CUTM/AGR/N0705: Herbaceous Plants or Hydrophytes in Aquaponics system

Description

This NOS focuses on the selection, care, and cultivation of herbaceous plants and hydrophytes in aquaponics systems. It includes understanding plant requirements, managing water quality, pest control, and harvesting techniques to ensure optimal plant health and yield while contributing to the overall balance and sustainability of the aquaponics environment.

Scope

The scope covers the following :

- The scope involves identifying suitable herbaceous/hydrophytic species, monitoring plant health, ensuring nutrient uptake, and managing environmental conditions. It includes planting, maintaining, and harvesting practices while ensuring sustainable, hygienic, and productive operations. These activities contribute to nutrient cycling and enhance the output of the aquaponics system.

Elements and Performance Criteria

Role of plants in Aquaponics system

To be competent, the user/individual on the job must be able to:

- PC1.** Choosing suitable plants for Aquaponics system.
- PC2.** Adjusting the aquaponic system to meet specific plant requirements.
- PC3.** Using companion planting strategies for pest control and nutrient enhancement.
- PC4.** Designing vertical systems for space efficiency.

Plant Nutrient Deficiencies

To be competent, the user/individual on the job must be able to:

- PC5.** Recognizing and addressing nutrient deficiencies in plants.
- PC6.** Strategies for preventing and managing plant diseases.
- PC7.** Understanding how plants respond to changes in water quality and conditions.
- PC8.** Adjusting the system to optimize plant growth.
- PC9.** Importance of oxygenation for healthy root systems.
- PC10.** Strategies for preventing and managing plant diseases.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the role of herbaceous and hydrophytic plants in nutrient absorption within aquaponics systems.
- KU2.** common herbaceous plants/hydrophytes suitable for aquaponics, such as lettuce, basil, mint, watercress, and spinach.

Qualification Pack

- KU3.** the growth requirements of these plants, including light, temperature, pH, and nutrient levels.
- KU4.** the root structures of hydrophytes and their adaptability to different aquaponics grow beds (NFT, media, DWC).
- KU5.** about pest management and disease prevention in water-based cultivation.
- KU6.** symptoms of nutrient deficiency or toxicity in plant leaves and roots.
- KU7.** planting, pruning, harvesting, and crop rotation techniques in aquaponics.
- KU8.** with organic, sustainable practices for plant health and productivity.
- KU9.** the importance of plant-microbe interactions and their role in water filtration.
- KU10.** compliance with food safety and hygiene standards for aquaponically grown edible herbs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Select appropriate herbaceous or hydrophytic plant species for different systems and seasons.
- GS2.** Monitor plant growth stages and identify any stress or abnormalities.
- GS3.** Maintain optimal water and environmental conditions for healthy plant growth.
- GS4.** Prune and harvest crops efficiently to maintain continuous production.
- GS5.** Handle seedlings and transplants with care to prevent damage.
- GS6.** Record plant growth parameters and production data systematically.
- GS7.** Manage pests organically and maintain hygiene in the plant area.
- GS8.** Use basic tools and equipment for planting and harvesting effectively.
- GS9.** Communicate with team members regarding crop health, schedules, and yield issues.
- GS10.** Follow safety and sustainability guidelines in day-to-day tasks.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Role of plants in Aquaponics system</i>	12	16	-	12
PC1. Choosing suitable plants for Aquaponics system.	3	4	-	3
PC2. Adjusting the aquaponic system to meet specific plant requirements.	3	4	-	3
PC3. Using companion planting strategies for pest control and nutrient enhancement.	3	4	-	3
PC4. Designing vertical systems for space efficiency.	3	4	-	3
<i>Plant Nutrient Deficiencies</i>	18	24	-	18
PC5. Recognizing and addressing nutrient deficiencies in plants.	3	4	-	3
PC6. Strategies for preventing and managing plant diseases.	3	4	-	3
PC7. Understanding how plants respond to changes in water quality and conditions.	3	4	-	3
PC8. Adjusting the system to optimize plant growth.	3	4	-	3
PC9. Importance of oxygenation for healthy root systems.	3	4	-	3
PC10. Strategies for preventing and managing plant diseases.	3	4	-	3
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0705
NOS Name	Herbaceous Plants or Hydrophytes in Aquaponics system
Sector	Agriculture
Sub-Sector	
Occupation	Cultivator
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

Qualification Pack

CUTM/VSQ/N7803: Maintain Health and Safety at Work Place.

Description

This NOS focuses on ensuring a safe and healthy working environment by promoting hazard identification, safe equipment handling, proper use of PPE, and adherence to workplace safety protocols. It involves assessing and mitigating risks, responding to emergencies, and minimizing environmental and workplace hazards to protect self, co-workers, and surroundings.

Scope

The scope covers the following :

- The scope of this NOS includes performing regular safety checks, using safety gear, handling hazardous materials responsibly, following emergency procedures, and maintaining hygiene and cleanliness in the workplace. It aims to cultivate a culture of safety and environmental responsibility while reducing risk, injury, and workplace accidents.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Undertaking basic safety checks before operation of all machinery and vehicles and hazards are reported to the appropriate supervisor
- PC2.** Work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy.
- PC3.** Reading and understanding the hazards of use and contamination mentioned on the labels of pesticides/fumigants etc
- PC4.** Assessing the risks prior to performing manual handling jobs and work according to currently recommended safe practice.
- PC5.** Use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** Dispose of waste safely and correctly in a designated area
- PC7.** Recognize risks to bystanders and take action to reduce risk associated with jobs in the workplace
- PC8.** Perform your work in a manner, which minimizes environmental damage all procedures and work instructions for controlling risk are followed closely.
- PC9.** Report any accidents, incidents or problems immediately to an appropriate person and take necessary immediate action to reduce further danger.
- PC10.** Follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC11.** Follow emergency procedures to company standard / workplace requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Basic safety checks and pre-operational inspection procedures for machinery and vehicles.
- KU2.** Importance of reporting hazards to supervisors immediately.
- KU3.** Identification and correct use of personal protective equipment (PPE).
- KU4.** Hazard symbols, warning labels, and safe handling instructions on pesticides/fumigants.
- KU5.** Risk assessment techniques for manual handling and other workplace tasks.
- KU6.** Safe usage, handling, and storage of tools, materials, and equipment.
- KU7.** Proper methods of waste segregation and disposal.
- KU8.** Identifying workplace risks to others and preventive action planning.
- KU9.** Environmental protection procedures and pollution control practices.
- KU10.** Emergency response protocols including fire, accident, and evacuation procedures.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Perform routine safety inspections and hazard identification confidently.
- GS2.** Demonstrate use of PPE according to task-specific needs.
- GS3.** Interpret chemical hazard labels and apply safety practices accordingly.
- GS4.** Conduct basic risk assessments for manual work operations.
- GS5.** Operate tools and equipment safely, and store them properly after use.
- GS6.** Segregate and dispose of waste in compliance with workplace guidelines.
- GS7.** Monitor surroundings to identify risks to co-workers and intervene appropriately.
- GS8.** Follow environmentally responsible practices during work execution.
- GS9.** Report incidents promptly and initiate basic emergency response actions.
- GS10.** Respond effectively to accidents and emergencies as per protocol.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	30	-	-
PC1. Undertaking basic safety checks before operation of all machinery and vehicles and hazards are reported to the appropriate supervisor	-	-	-	-
PC2. Work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy.	2	3	-	-
PC3. Reading and understanding the hazards of use and contamination mentioned on the labels of pesticides/fumigants etc	2	3	-	-
PC4. Assessing the risks prior to performing manual handling jobs and work according to currently recommended safe practice.	2	3	-	-
PC5. Use equipment and materials safely and correctly and return the same to designated storage when not in use	2	3	-	-
PC6. Dispose of waste safely and correctly in a designated area	2	3	-	-
PC7. Recognize risks to bystanders and take action to reduce risk associated with jobs in the workplace	2	3	-	-
PC8. Perform your work in a manner, which minimizes environmental damage all procedures and work instructions for controlling risk are followed closely.	2	3	-	-
PC9. Report any accidents, incidents or problems immediately to an appropriate person and take necessary immediate action to reduce further danger.	2	3	-	-
PC10. Follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Follow emergency procedures to company standard / workplace requirements	2	3	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7803
NOS Name	Maintain Health and Safety at Work Place.
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	21/10/2027
NSQC Clearance Date	22/10/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

Qualification Pack

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
 - Questions are mapped to the specified assessment criteria.
 - Assessor must be ToA certified & trainer must be ToT Certified
4. Types of evidence or evidence-gathering protocol:
- Time-stamped & geotagged reporting of the assessor from assessment location.
 - Centre photographs with signboards and scheme specific branding
 - Video Call verification/online proctoring will be done during the assessment from AB.
5. Method of verification or validation:
- Surprise visit to the assessment location
6. Minimum Percentage Required
- 70% overall to pass.
 - Minimum 70% in each NOS
7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.
8. Assessor Details: ToA Certified Assessor with relevant experience

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 60

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N0701.Introduction to Aquaponics	30	40	-	30	100	10

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N0702.Aquaponics System Layout and its Components.	40	60	-	-	100	20
CUTM/AGR/N0703.The Nitrogen cycle/Water testing and Quality control	40	60	-	-	100	20
CUTM/AGR/N0704.Ornamental fish in Aquaponics system	30	70	-	-	100	20
CUTM/AGR/N0705.Herbaceous Plants or Hydrophytes in Aquaponics system	30	40	-	30	100	10
CUTM/VSQ/N7803.Maintain Health and Safety at Work Place.	20	30	-	-	50	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	210	330	-	60	600	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.